Alejandro **Fontan Villacampa**

Research Scientist | 3D Computer Vision & Visual SLAM

Personal Website

About me

I'm a Research Scientist with expertise in Computer Vision, Robotics, AR/VR, and Deep Learning. Passionate about mentorship and collaboration. My career is dedicated to exploring and shaping the future of Visual SLAM and Localization.

Areas of specialization

Visual SLAM • Localization • Autonomous Navigation • Visual Odometry • VPR

Languages

Spanish (native)
English (IELTS 8.0)

Links

GitHub
LinkedIn
Scholar
@AFontanVillcmp

References

Michael Milford (QUT)
Javier Civera (Univ. Zaragoza)
Tobias Fischer (QUT)
Rudolph Triebel (DLR)
Markus Eder (Wikitude)

Other Activities

First Aid Course & CPR
Professional Music Studies

Hobbies & Interests

Scuba Diving • Piano • Trail Running • Films

Fun Fact

I keep trying to run monocular SLAM on GoPro footage from my scuba diving trips.

Short Resumé

2022–pres. **Research Scientist | 3D Computer Vision & Visual SLAM**
Queensland University of Technology · Brisbane, Australia
At the QCR, I lead research on Visual SLAM. Recent highlights include *AllFeature-VSLAM*, a complete framework that tightly integrates diverse features, and a novel *ground-truth-free tuning* method for SLAM/SfM. I also mentor PhD students and our research team in Localization tasks. QUT

2018–2022 **PhD Researcher (Information-Driven Navigation)**
German Aerospace Center (DLR) · Munich, Germany
Pioneered *Information-driven direct RGB-D odometry* and *Semi-direct SID-SLAM* to improve visual odometry for mobile and aerial robots in planetary exploration. Deployed and evaluated on a custom DLR multicopter, contributing to its development as a research platform. DLR

2017–2018 **Internship as Software Engineer**
Wikitude GmbH · Salzburg, Austria
Developed a computer vision AR library for mobile OS like Android and iOS, focusing on implementing and evaluating egocentric visual-inertial SLAM algorithms on mobile devices, *OKVIS* and *VINS-Mono*. Created calibration solutions for phone cameras and IMUs' extrinsics. Wikitude Webpage

Latest Projects

VSLAM-LAB: comprehensive framework for Visual SLAM baselines & datasets.
AllFeature-VSLAM: tightly coupled multi-feature monocular SLAM.
Bommie Toolkit: automated 3D reconstruction for coral reefs.
AnyFeature-VSLAM: automates using any chosen feature in Visual SLAM.

Workshop Organizer:
Mapping-the-Reef@RSS26 **(1st Org.)**
The-Good-Reviewer@ICRA26 **(1st Org.)**
Unifying-SLAM@RSS25 **(1st Org.)**

Education & Teaching

2026 **Lecturer for ENN584**
SLAM-Course ·

2025 **CV for Spatial Intelligence**
ICVSS · Summer School

2018–22 **PhD System Engineering and Programming**
Univ. Zaragoza · DLR
Advisor: Dr. Javier Civera
Co-Advisor: Dr. Rudolph Triebel

2011–17 **Degree & Master of Industrial Engineering**
EINA · Univ. Zaragoza

Main Publications

ICUAS 26	*Aerial Visual Place Recognition in Antarctica.*
arXiv 25	*Pixi: Unified Software Development and Distribution for Robotics.*
IROS 25	*VSLAM-LAB: A Comprehensive Framework for Visual SLAM Methods and Datasets.*
arXiv 25	*Look Ma, No Ground Truth! Ground-Truth-Free Tuning of SfM and VSLAM.*
IROS 25	*Image-Based Relocalization and Alignment for Long-Term Monitoring of Dynamic Underwater Environments.*
RSS 24	*AnyFeature-VSLAM: Automating the Usage of Any Feature into Visual SLAM.*
ICRA 24	*Adaptive Outlier Thresholding for Bundle Adjustment in VSLAM.*
BMVC 23	*Motion-Bias-Free Feature-Based SLAM.*
RA-L 23	*SID-SLAM Semi-Direct Information Driven RGB-D SLAM.*
RA-L 22	*A Model for Multi-View Residual Covariances based on Perspective Deformation.*
ICRA 21	*DOT: Dynamic Object Tracking.*
JFR 21	*The MADMAX data set for visual-inertial rover navigation on Mars.*
CVPR 20 Oral	*Information-Driven Direct RGB-D Odometry.*

Programming

C++ / Python **PyTorch / CUDA** **OpenCV / Rerun** **ONNX / Tensor-RT**
Anaconda / Pixi **ROS1 / ROS2** **Ceres / g2o / GTSAM** **Matlab / LaTeX / Git**

Data Structures & Algorithms — 179 LeetCode problems solved (73 Easy · 101 Medium · 5 Hard)

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QUT Brisbane, Australia

Publications — 398 citations · h-index 8 (Google Scholar, Aug. 2026)

ICUAS 26	*Aerial Visual Place Recognition in Antarctica: Towards Robust Monitoring in Extreme Environments.*
arXiv 26	*Long-Term Multi-Session 3D Reconstruction Under Substantial Appearance Change.*
ICRA 26	*Event-LAB: Towards Standardized Evaluation of Neuromorphic Localization Methods.*
WACV 26	*Robust Scene Coordinate Regression via Geometrically-Consistent Global Descriptors.*
ICRAW 26	*Visual SLAM on a Lunar Testbed for Polarization Cameras Under Challenging Lighting Conditions.*
arXiv 25	*Pixi: Unified Software Development and Distribution for Robotics & AI.*
IROS 25	*VSLAM-LAB: A Comprehensive Framework for Visual SLAM Methods and Datasets.*
IROS 25	*Image-Based Relocalization and Alignment for Long-Term Monitoring of Dynamic Underwater Environments.*
ICCVW 25	*FUSELOC: Fusing Global and Local Descriptors for Fast and Robust 2D-3D Matching in Visual Localization.*
arXiv 25	*Event-Based Visual Teach-and-Repeat via Fast Fourier-Domain Cross-Correlation.*
arXiv 25	*Look Ma, No Ground Truth! Ground-Truth-Free Tuning of Structure from Motion and Visual SLAM.*
WACV 24	*FocusTune: Tuning Visual Localization through Focus-Guided Sampling.* (32 citations)
RSS 24	*AnyFeature-VSLAM: Automating the Usage of Any Chosen Feature into Visual SLAM.* (14 citations)
ICRA 24	*Adaptive Outlier Thresholding for Bundle Adjustment in Visual SLAM.*
RA-L 24	*Forward Prediction of Target Localization Failure through Pose Estimation Artifact Modelling.*
RA-L 23	*SID-SLAM: Semi-Direct Information-Driven RGB-D SLAM.* (20 citations)
BMVC 23	*Motion-Bias-Free Feature-Based SLAM.*
RA-L 22	*A Model for Multi-View Residual Covariances based on Perspective Deformation.*
ICRA 21	*DOT: Dynamic Object Tracking for Visual SLAM.* (133 citations)
JFR 21	*The MADMAX Data Set for Visual-Inertial Rover Navigation on Mars.* (92 citations)
IAC 21	*Preliminary Results for the Multi-Robot, Multi-Partner, Multi-Mission, Planetary Exploration Analogue Campaign on Mount Etna.* (17 citations)
CVPR 20	*Information-Driven Direct RGB-D Odometry.* **Oral presentation.** (41 citations)

Open-Source Software

VSLAM-LAB (★ 500+ · ⑂ 69) **C++ / Python** **Anaconda / Pixi** **ROS1 / ROS2** — Unified, Pixi-based framework to compile, run, and evaluate 13+ Visual SLAM baselines (ORB-SLAM2/3, DROID-SLAM, MASt3R-SLAM, AllFeature/AnyFeature-VSLAM, DPVO, MonoGS, VGGT-SLAM, PyCuVSLAM, ...) across 38+ datasets from a single command line; per-baseline Pixi environments isolate conflicting CUDA/dependency stacks, and a config-driven pipeline standardizes dataset formatting, execution, and evaluation for one-command, reproducible benchmarks. Lead developer.

@VSLAM-LAB tutorials & demos: Intro @ IROS 25 · Setup in under 2 min · AllFeature / DROID / MASt3R-SLAM in Iceland · PyCuVSLAM on ROVER

Self-Supervised Deep SLAM Fine-Tuning **PyTorch / CUDA** — Fine-tuning DROID-SLAM's ConvGRU update operator (global-context gate, correlation volume) inside its differentiable SE3 Bundle Adjustment layer, on real sensor data; fixed a bug where missing-depth pixels ($\leq$64% of a frame on real sensors) were silently treated as confident supervision. Builds on our ground-truth-free tuning methodology (*Look Ma, No Ground Truth!*), replacing mocap ground truth with an image-perturbation sensitivity signal that correlates with true ATE — enabling self-supervised fine-tuning on unlimited unlabeled video. Cut mean ATE by ~45% on ETH3D-SLAM.

AllFeature-VSLAM **C++ / Python** **Ceres / g2o / GTSAM** **ONNX / Tensor-RT** — Real-time monocular SLAM that tightly couples heterogeneous binary and learned feature descriptors (ORB, AKAZE, BRISK, SIFT, KAZE, ALIKED, SuperPoint) within a single joint map and bundle adjustment, instead of committing to one feature per pipeline; TensorRT-accelerated LightGlue matching for learned features. Outperforms ORB-SLAM2/3 and deep-learned VO (DROID-VO, DPVO) on EuRoC and TUM-RGBD. Lead developer.

@VSLAM-LAB demos: AllFeature / DROID / MASt3R-SLAM in Iceland · Marine Conservation · on the MADMAX Data Set

Bommie Toolkit **OpenCV / Rerun** — End-to-end pipeline for long-term coral reef monitoring: stereo rig calibration (Kalibr), video-to-COLMAP structure-from-motion with SAM-based masking, and Gaussian Splatting reconstruction (nerfstudio), with refraction removal for underwater capture. Built with Pixi; co-developed with Emilio Olivastri.

Mentorship & Supervision

Chelim Lim — PhD, principal supervisor (2026–pres.), QUT. *Adaptive Refractive Visual SLAM for Underwater Robotics.*

Krishnan Harikumar — PhD, associate supervisor (2025–pres.), QUT. *Compact and Multimodal 3D Reconstruction of Large-Scale Environments Using Gaussian Splatting.*

Beverley Gorry — PhD, principal supervisor (2024–pres.), QUT. *Applying Robotic Vision for Revisiting the Reef.* (IROS 25)

Son Tung Nguyen — PhD, associate supervisor (2022–2025), QUT. *Memory-Efficient Visual Localization.* (WACV 24, ICCVW 25, WACV 26)

Irene Ballester Campos — MSc thesis, associate supervisor (2020), DLR. *Robust Visual SLAM in Dynamic Environments Combining Deep Learning and Multiview Consistency.* (ICRA 21)

SLAM-Course — Lecturer, QUT ENN584 (2026–pres.). Open lecture slides (Factor Graphs, Photogrammetry) and hands-on Python exercises (EKF-SLAM, GTSAM/g2o, Photogrammetry), including a factor-graph pipeline that estimates trajectories from real iPhone-collected ArUco marker data.

Academic Service

Associate Editor: IROS (2024–2026).

Workshop organization (1st organizer): Mapping-the-Reef (RSS 2026), The-Good-Reviewer (ICRA 2026), Unifying-SLAM (RSS 2025).

Journal reviewer: IEEE RA-L (44+ reviews), IEEE T-RO (17+ reviews).

Conference reviewer: RSS, CVPR, ICCV, ECCV, ICRA, IROS.

Recognition: CVPR 2020 Oral presentation.

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